

Insufficient Evidence for Biological Growth Cessation Mechanisms in Provided Physics and Computer Science Literature

Input Question

Why do we stop growing?

Domain

Biology

Validation Status

needs_data

Problem Statement

The user asks for the biological mechanism behind the cessation of physical growth, citing a booklet claim that genes determine this timing upon reaching reproductive maturity. However, the provided evidence set consists entirely of documents from computer science (process algebra, computer vision) and high-energy physics (particle physics), where the term 'stop' refers to computational states, traffic signs, or subatomic particles (stop squarks), not physiological development.

Rationale

Scientific integrity requires that answers be grounded in provided evidence. The allowed evidence IDs (EV-Q039-42560d81e95b0c4f23de71a3, EV-Q039-ff67ff8ae23635737ef4b0d0, EV-Q039-1bc8d552959bd3bbb7c80972, EV-Q039-86b487e1058c423e23ac0f6d) are semantically irrelevant to human biology. Therefore, the only scientifically valid response is to demonstrate this domain mismatch and declare insufficient evidence for the biological question, rather than hallucinating biological mechanisms from unrelated physics/CS texts.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Domain Mismatch Null Hypothesis: The allowed evidence set contains zero information regarding biological growth cessation, rendering the question scientifically unanswerable within current constraints.
- **Mechanism**: The semantic content of all allowed evidence IDs (EV-Q039-42560d81e95b0c4f23de71a3, EV-Q039-ff67ff8ae23635737ef4b0d0, EV-Q039-1bc8d552959bd3bbb7c80972, EV-Q039-86b487e1058c423e23ac0f6d) is strictly confined to non-biological domains (process algebra, particle physics, computer vision). The term 'stop' functions exclusively as a technical operator or physical object label in

these texts, with no causal or descriptive link to physiological development. Therefore, no valid inference about human growth can be derived.

- ****Falsifiable Prediction****: If exact string matching for a predefined list of biological keywords (e.g., 'epiphysis', 'growth plate', 'hormone', 'skeletal', 'chondrocyte') against the full text of any allowed evidence ID yields >0 matches, this hypothesis is falsified.

- ****Required Observations****: Exact string match count of biological keywords in EV-Q039-42560d81e95b0c4f23de71a3 equals 0 ; Exact string match count of biological keywords in EV-Q039-ff67ff8ae23635737ef4b0d0 equals 0 ; Exact string match count of biological keywords in EV-Q039-1bc8d552959bd3bbb7c80972 equals 0 ; Exact string match count of biological keywords in EV-Q039-86b487e1058c423e23ac0f6d equals 0

- ****Risk of Being Wrong****: Low. The quoted excerpts and titles explicitly define non-biological contexts. Risk exists only if obscure biological metaphors appear in unquoted sections, which is statistically negligible given the specialized nature of the source material.

Hypothesis 2

- ****Hypothesis****: Polysemous Retrieval Artifact Hypothesis: The presence of physics/CS evidence for a biology query is an artifact of lexical ambiguity where 'stop' was matched without domain disambiguation, confirming that the current evidence set is invalid for biological inquiry.

- ****Mechanism****: The retrieval system indexed documents based on the token 'stop' present in EV-Q039-42560d81e95b0c4f23de71a3 (STOP α VMS process), EV-Q039-ff67ff8ae23635737ef4b0d0 (stop squark), and EV-Q039-1bc8d552959bd3bbb7c80972 (stop signs). This mechanism explains the domain mismatch as a systematic retrieval failure rather than a lack of literature in the broader universe, but confirms insufficiency within the allowed set.

- ****Falsifiable Prediction****: If context analysis shows that the token 'stop' in any allowed evidence ID appears in a sentence containing biological entities or developmental processes, the artifact hypothesis is weakened (suggesting potential relevance). If 'stop' appears only in technical/non-biological contexts, the artifact hypothesis is supported.

- ****Required Observations****: Contextual classification of every instance of 'stop' in EV-Q039-42560d81e95b0c4f23de71a3 as 'process_algebra' ; Contextual classification of every instance of 'stop' in EV-Q039-ff67ff8ae23635737ef4b0d0 as 'particle_physics' ; Contextual classification of every instance of 'stop' in EV-Q039-1bc8d552959bd3bbb7c80972 as 'computer_vision' ; Verification that no instance of 'stop' co-occurs with biological terms in allowed IDs

- ****Risk of Being Wrong****: Medium. While highly likely given the titles, this hypothesis relies on interpreting the *absence* of biological context as proof of retrieval error. It is secondary to the Null Hypothesis because it diagnoses the system rather than the science.

Technical Details

This experiment validates the 'Domain Mismatch Null Hypothesis' by rigorously demonstrating that the allowed evidence set contains no biological information regarding human growth cessation. The methodology relies on deterministic string matching and keyword density analysis within the allowed evidence IDs, avoiding dependence on external embeddings or ontologies for the primary falsification test. The core technical approach involves: 1) Extracting full-text content from the four allowed arXiv PDFs (EV-Q039-42560d81e95b0c4f23de71a3,

EV-Q039-ff67ff8ae23635737ef4b0d0, EV-Q039-1bc8d552959bd3bbb7c80972, EV-Q039-86b487e1058c423e23ac0f6d). 2) Performing exact case-insensitive string matching for a predefined list of high-specificity biological keywords (e.g., 'epiphysis', 'chondrocyte', 'somatotropin', 'osteoblast', 'puberty') against the extracted text. 3) Analyzing the contextual usage of the token 'stop' to confirm its exclusive association with non-biological domains (process algebra, particle physics, computer vision). A result of zero biological keyword matches and 100% non-biological context classification for 'stop' confirms the hypothesis.

Datasets

Source

```
```json
[
{
 "id": "EV-Q039-42560d81e95b0c4f23de71a3",
 "description": "Full text of arXiv:1908.06601 regarding Vending Machine Systems (VMS) and process algebra."
},
{
 "id": "EV-Q039-ff67ff8ae23635737ef4b0d0",
 "description": "Full text of arXiv:1212.6847 regarding stop squark and neutralino coannihilation in high-energy physics."
},
{
 "id": "EV-Q039-1bc8d552959bd3bbb7c80972",
 "description": "Full text of arXiv:1710.03337 regarding adversarial examples on physical stop signs in computer vision."
},
{
 "id": "EV-Q039-86b487e1058c423e23ac0f6d",
 "description": "Full text of arXiv:1401.7989 regarding R-Parity Breaking and Stop LSP Decays in MSSM."
}
]
```
```

Target

```

"""json

{

"description": "External Validation Resource: A control corpus of peer-reviewed articles from PubMed/PMC
explicitly discussing 'human epiphyseal fusion' and 'growth hormone regulation'. This dataset is strictly used for
external validation of the semantic gap and is NOT part of the allowed evidence set. It serves only to establish a
positive control for what biological relevance looks like, ensuring the negative result in the source set is
meaningful.",

"source_type": "external_validation_resource"

}

"""

```

Paper Abstract

Background: The question 'Why do we stop growing?' implies a biological mechanism involving genetic regulation and skeletal maturation. However, the provided evidence set consists of arXiv papers from unrelated fields. Methods: We performed a deterministic keyword analysis on four allowed evidence IDs (EV-Q039-42560d81e95b0c4f23de71a3, EV-Q039-ff67ff8ae23635737ef4b0d0, EV-Q039-1bc8d552959bd3bbb7c80972, EV-Q039-86b487e1058c423e23ac0f6d) searching for biological terms such as 'epiphysis', 'chondrocyte', and 'puberty'. We also analyzed the contextual usage of the token 'stop'. Validation Plan: The hypothesis predicts zero biological keyword matches and 100% non-biological context classification for 'stop'. Results: pending (待执行验证实验). Conclusion: The available evidence is insufficient to answer the biological question due to severe domain mismatch.

Methods

1. **Text Extraction**: Use pdfminer.six to extract raw text from the four source PDFs identified by their evidence IDs.
2. **Deterministic Keyword Matching**: Define a strict list of biological terms: ['epiphysis', 'growth plate', 'chondrocyte', 'osteoblast', 'somatotropin', 'gonadotropin', 'puberty', 'skeletal maturation']. Perform case-insensitive exact string matching against the full text of each source document.
3. **Contextual Classification of 'Stop'**: Extract all sentences containing the token 'stop'. Manually or via simple rule-based classification to label each instance as 'non-biological'.
4. **Negative Control Verification**: Confirm that the count of biological keywords in all source documents is exactly zero.

Experiments

Baselines

```

"""json

[

"Null Baseline: A document known to contain no text (empty string) to verify the keyword matcher returns zero
matches correctly.",

```

"Random Noise Baseline: A document filled with random English words to ensure the keyword matcher does not produce false positives on non-sensical text."

]

...

Metrics

```json

[

"Biological Keyword Hit Rate: The total count of exact matches for the predefined biological keyword list across all allowed evidence IDs. Expected value: 0.",

"Non-Biological Context Precision: The proportion of 'stop' token instances classified as non-biological. Expected value: 1.0 (100%).",

"Evidence Relevance Confirmation Score: A binary metric (0 or 1) indicating whether any sentence in the source documents contains both the token 'stop' and a biological entity. Expected value: 0."

]

...

## Ablation

Remove the case-insensitivity flag to test if capitalization variations (e.g., 'Epiphysis' vs 'epiphysis') affect the zero-match result. This ensures the robustness of the negative finding.

## Validation Protocol

Manual audit of all extracted text segments containing the word 'stop' by two independent reviewers to confirm they do not refer to biological growth. Inter-reviewer agreement must be 100% for the 'non-biological' classification to validate the automated string matching results.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- \*\*EV-Q039-42560d81e95b0c4f23de71a3\*\* · arxiv · arXiv:1908.06601
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1908.06601.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:2|section:page-2|paragraph:1;

content\_sha256=222f76ea6b59922aaa200907fcbfbf677a8781246559cd72f5eb9356dcfb5f2f

- **\*\*EV-Q039-ff67ff8ae23635737ef4b0d0\*\*** · arxiv · arXiv:1212.6847
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1212.6847.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:3|section:page-3|paragraph:1;  
content\_sha256=d5fb305f1d48557cf2fe8dfdc11c83f9475a993e400900c1e56f005ac433ad33
- **\*\*EV-Q039-1bc8d552959bd3bbb7c80972\*\*** · arxiv · arXiv:1710.03337
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1710.03337.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=28e6aae88c6676ba0bc0767ebebe77bff4a97a584f33d35694fb5be05223cf1f
- **\*\*EV-Q039-86b487e1058c423e23ac0f6d\*\*** · arxiv · arXiv:1401.7989
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1401.7989.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=b461750a4625ceda97ae7c8a21274da127de83e41bac8b2972077a29b215136e

## Reviewer Comments

- The revised plan successfully addresses all critical issues from the previous review iteration.
- The target dataset is now correctly labeled as 'external\_validation\_resource' and explicitly distinguished from allowed evidence, resolving the evidence grounding violation.
- Hypothesis 1 has been appropriately demoted to a secondary diagnostic role, with the Domain Mismatch Null Hypothesis established as the primary scientific output.
- The experiment design now includes a deterministic string-matching metric ('Evidence Relevance Confirmation Score') that does not depend on external embeddings, ensuring robustness and reproducibility.
- The reproducibility checklist correctly clarifies that external biomedical ontologies are optional dependencies for validation only, not required for the primary falsification test.
- All supporting evidence IDs in the recommended hypothesis are valid and present in the allowed\_evidence\_ids list.
- Results field correctly remains 'pending' with no fabrication of experimental outcomes.

## Revision History

- auto\_revision\_1: 依据评审意见重做假设与实验设计。

## Reproducibility Checklist

- Verify access to the four specific arXiv PDFs via their provided URLs in the evidence catalog.
- Confirm the exact list of biological keywords used for matching is logged and fixed before execution.
- Ensure the text extraction script handles PDF encoding correctly to avoid missing hidden text.
- Clarify that access to external biomedical ontologies (UMLS/MeSH) is NOT required for the primary string-matching experiment, only for the optional external validation resource.
- Log all keyword match counts and context classifications in a structured CSV for audit.